ARRAYS/STRINGS

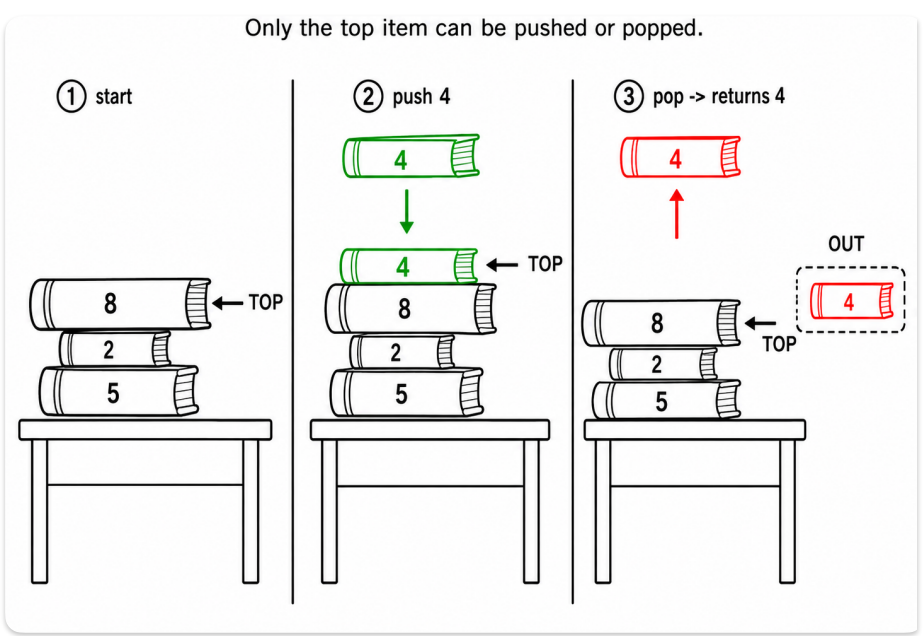
Two pointers
Sliding window
Prefix sum

HASHING
(SETS & DICTIONARIES)

Anagram grouping
Frequent maps
Counting

When to think of it

- I need to restore the most recently saved state first.
- I need undo / backtracking behavior.
- I need to return to the latest unfinished or previous context.
- The natural processing order is last-in, first-out.

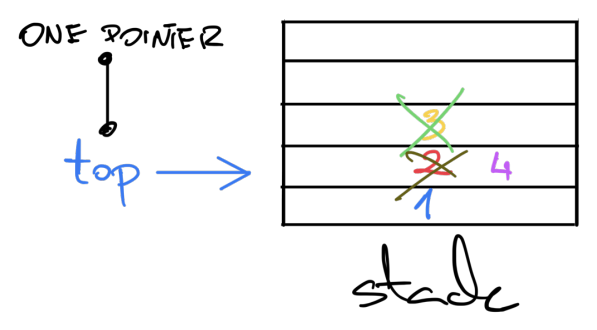

Only the top item can be pushed or popped.
① start
② push 4
③ pop -> returns 4
8
2
5
4
TOP
OUT

O(n)
BOTTOM ITEM
remove
insert

O(1)
TOP ITEM
change
append(x) push onto
pop() remove
stack[-1] read
BOTTOM ITEM
read stack[0]
LEN()

STACKS/QUEUES

STACK
LIFO


ONE POINTER
top
1
2
3
4
stack

push(1)
push(2)
push(3)
pop()
pop()
push(4)

Python
list []

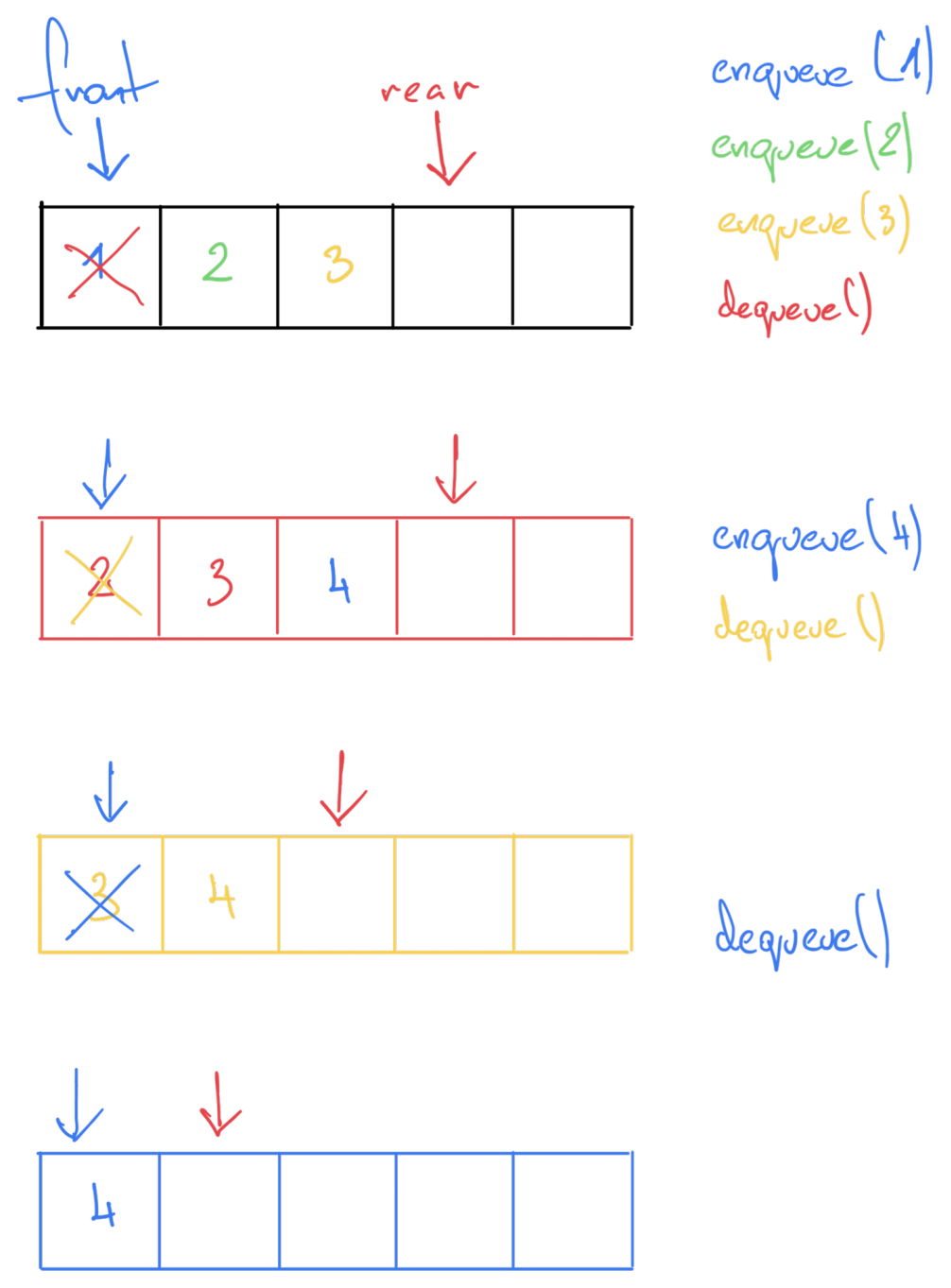
USE CASES
backtracking
find correct path through maze
depth-first search
undo (pop) / redo (push)

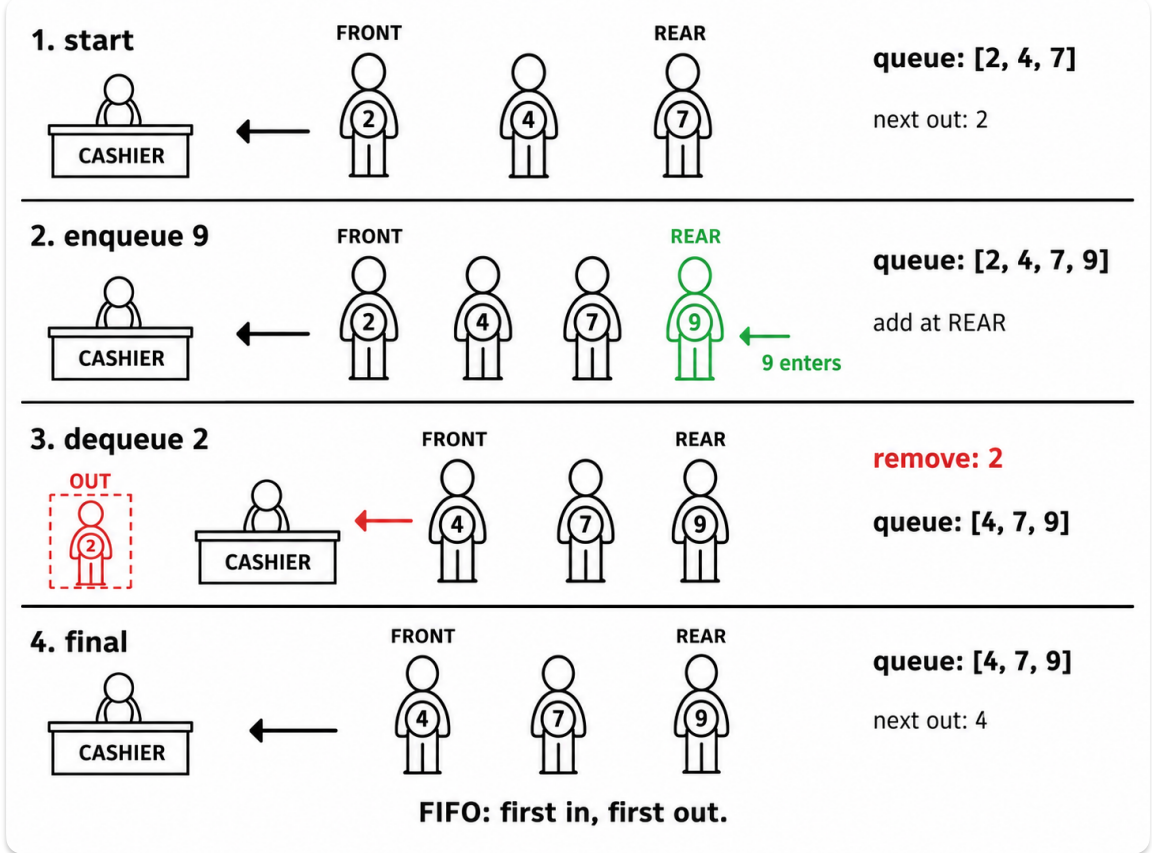
newest element
processed first

QUEUE
FIFO

O(1)
FRONT
queue.popleft() remove
read
BACK
add queue.append(x)
read
LEN()

Python
from collections import deque


front
rear
1
2
3
enqueue(1)
enqueue(2)
enqueue(3)
dequeue()
2
3
4
enqueue(4)
dequeue()
3
4
dequeue()
4

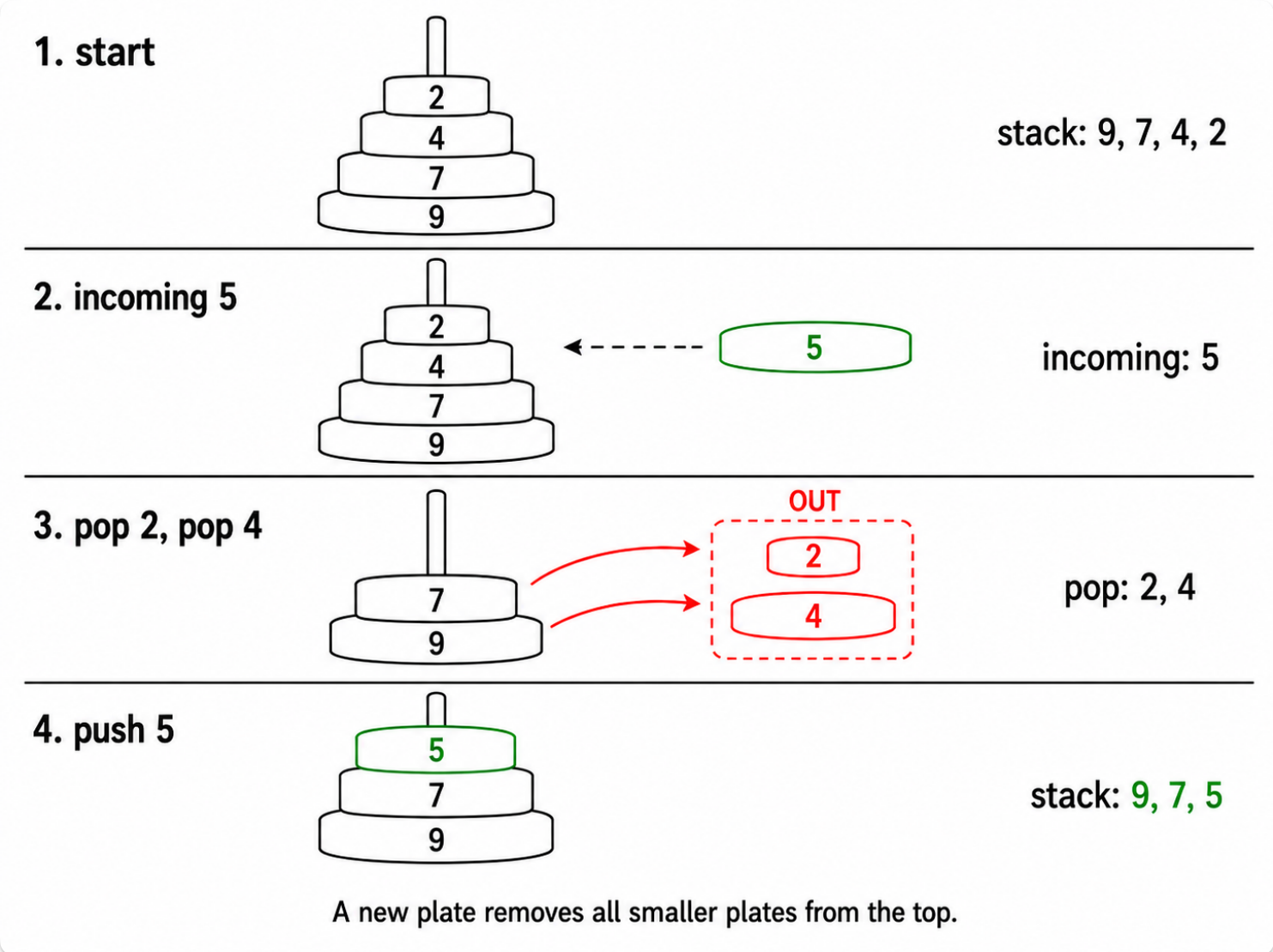

1. start
FRONT
REAR
CASHIER
2
4
7
queue: [2, 4, 7]
next out: 2
2. enqueue 9
FRONT
REAR
CASHIER
2
4
7
9
9 enters
queue: [2, 4, 7, 9]
add at REAR
3. dequeue 2
OUT
2
CASHIER
FRONT
REAR
4
7
9
remove: 2
queue: [4, 7, 9]
4. final
FRONT
REAR
CASHIER
4
7
9
queue: [4, 7, 9]
next out: 4
FIFO: first in, first out.

USE CASES
operating systems
- CPU and disk scheduling
Spotify
- playlist
breath-first search

oldest element processed first

When to think of it

- The problem asks for the **first / nearest later value** that is greater or smaller.
- I scan left to right while some previous values are still **waiting for an answer**.
- One new value can resolve **several previous values at once**.
- Once a value gets its answer, I can remove it permanently.


1. start
2
4
7
9
stack: 9, 7, 4, 2
2. incoming 5
2
4
7
9
5
incoming: 5
3. pop 2, pop 4
7
9
OUT
2
4
pop: 2, 4
4. push 5
5
7
9
stack: 9, 7, 5
A new plate removes all smaller plates from the top.

MONOTHONIC STACK

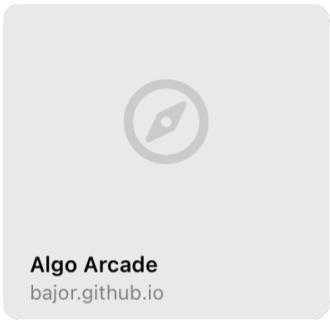
VIDEO

MONO STACK

2.9K likes, 90 comments. "Monotonic Stack Data Structure Explained"

AlgoMonster
youtu.be

GAME


Algo Arcade
bajor.github.io